

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

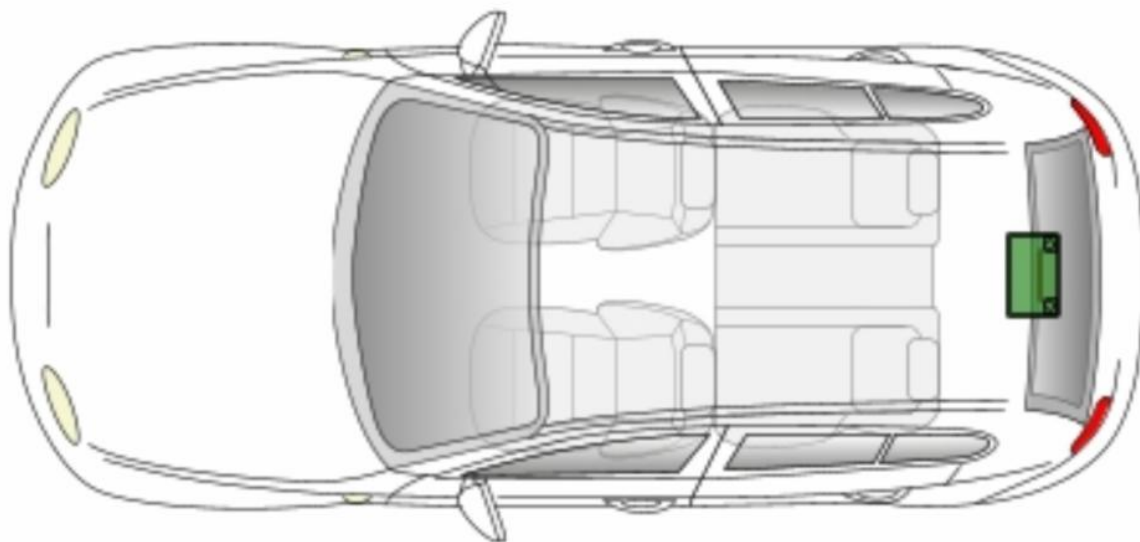
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

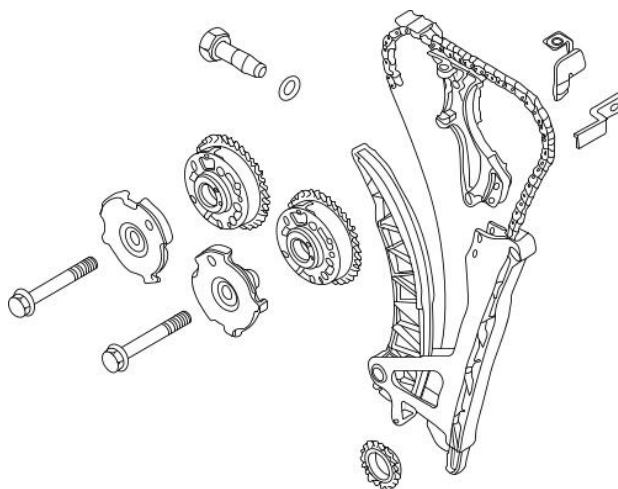
Battery location



Luggage compartment

Timing chain layout

Repair Manuals



Removal

Disconnect the battery earth cable

Remove the mass airflow meter

Remove the air filter housing

Remove the ignition coil cover

Remove the ignition coils

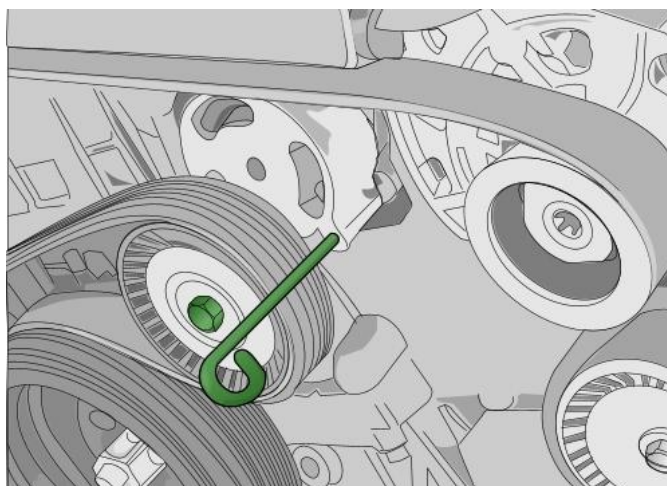
Remove the spark plugs

Remove the cylinder head cover

Turn the tensioner body wheel to release the belt tension

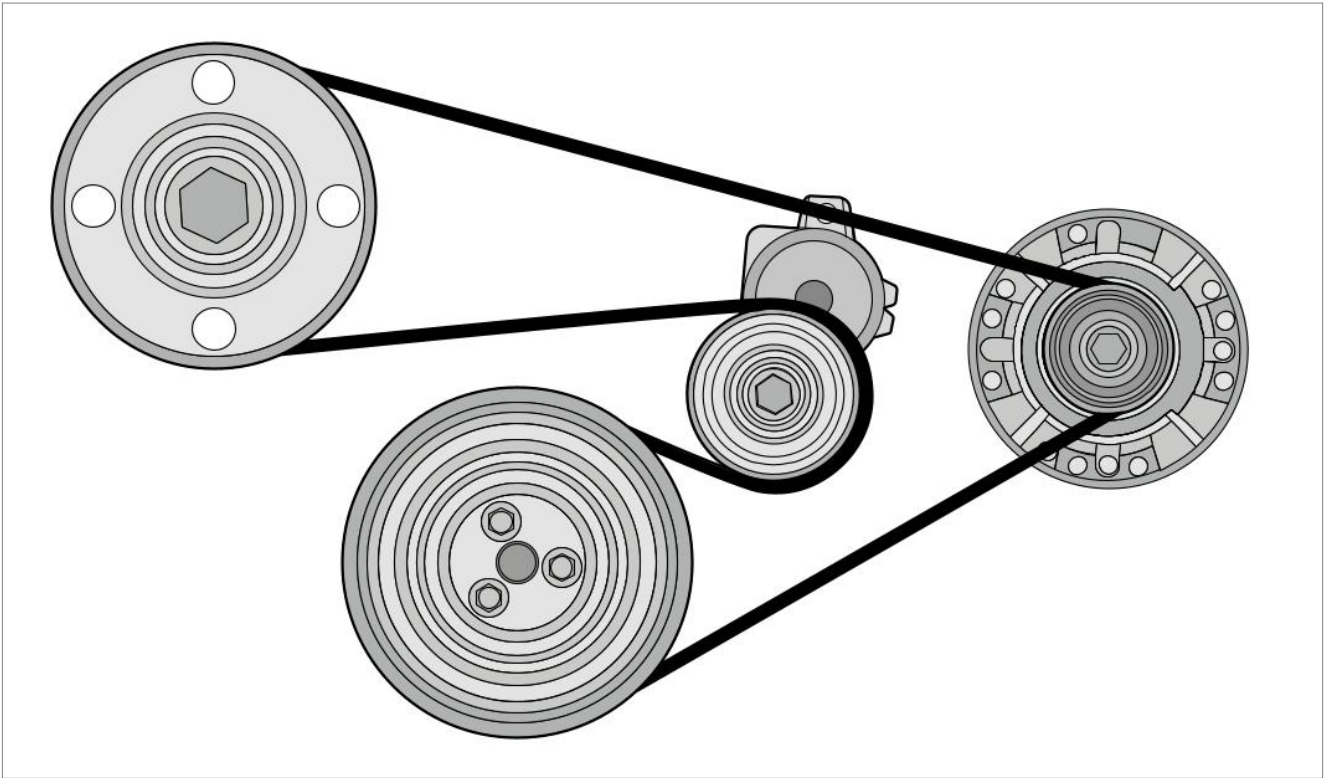
Lock the tensioner; use a locking pin (11 3 340)

Remove the ancillary drive belt

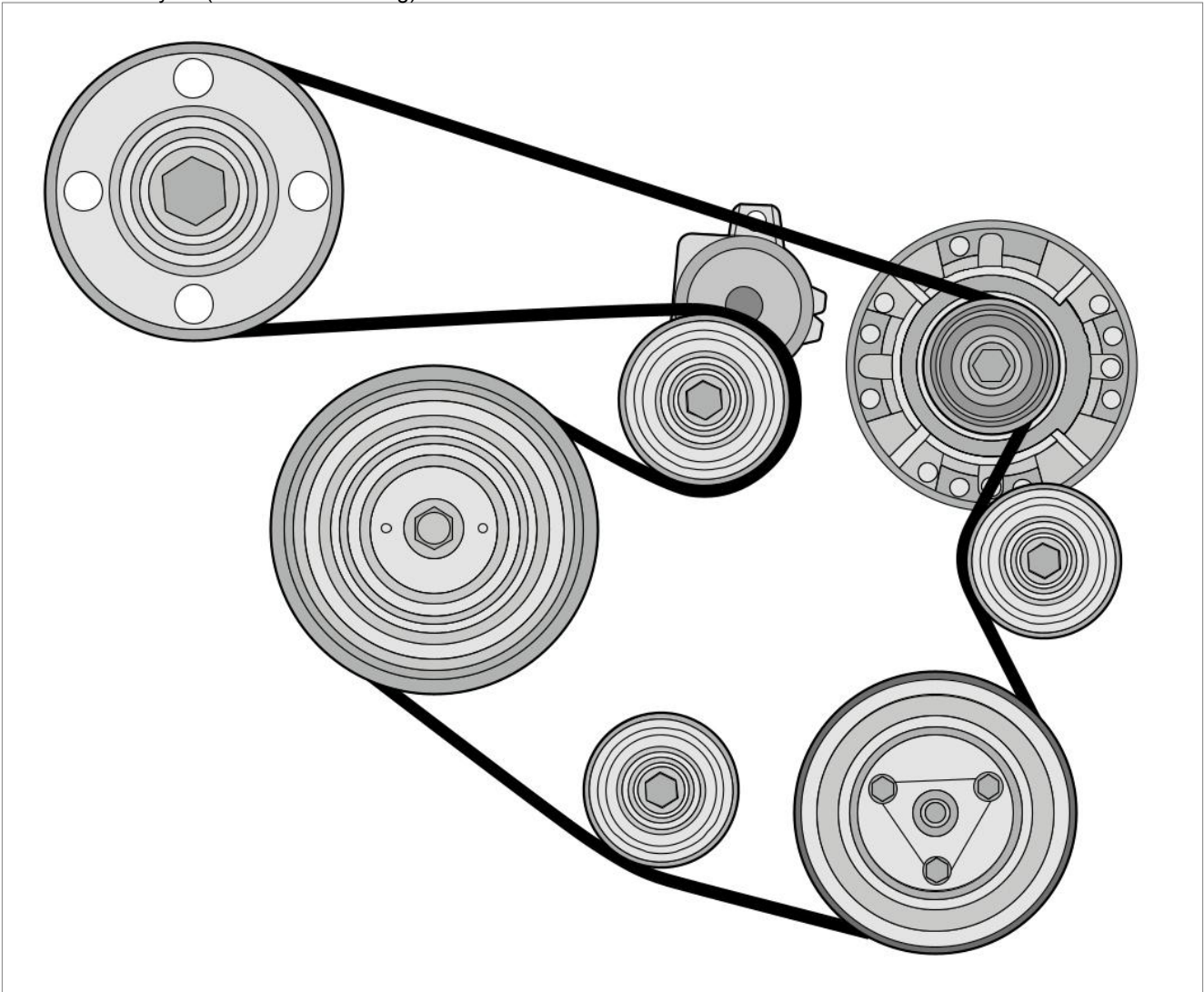


Drive belt layout (Without air conditioning)

Repair Manuals



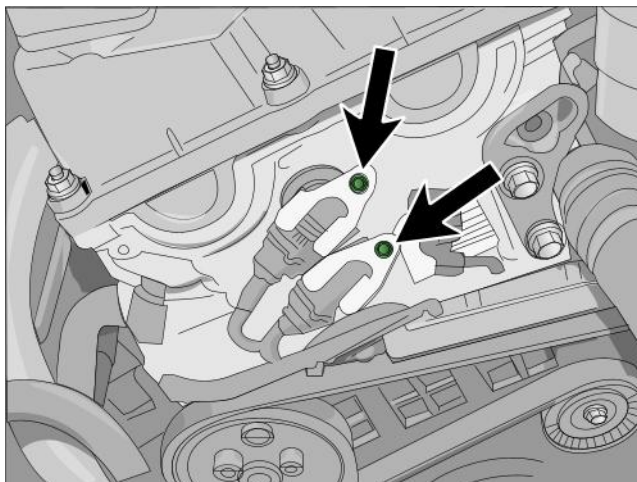
Drive belt layout (With air conditioning)



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Remove the vibration damper

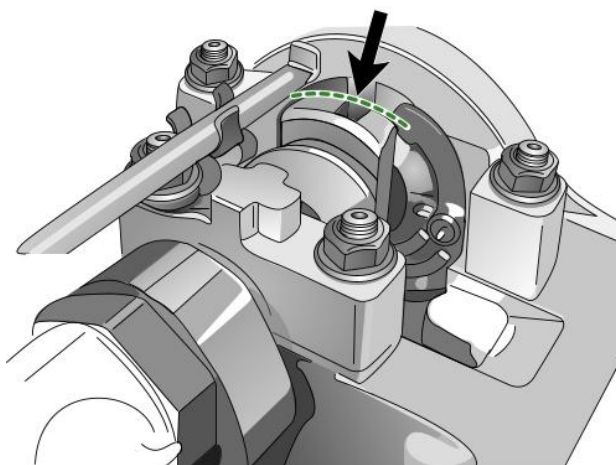
Disconnect the connector(s) from the solenoid valve



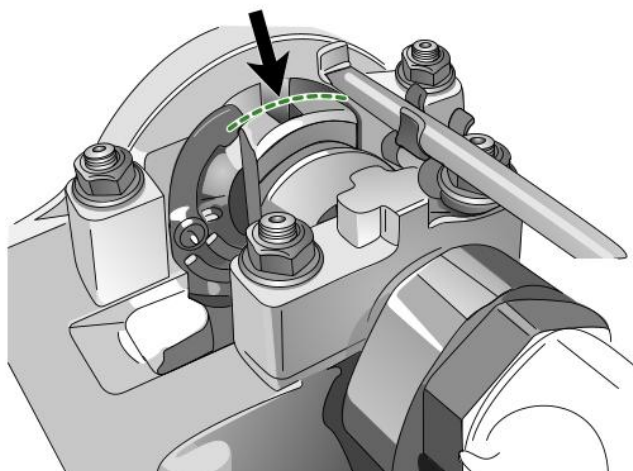
Turn the crankshaft to TDC for cylinder 1

Check the position of the camshafts

Inlet:

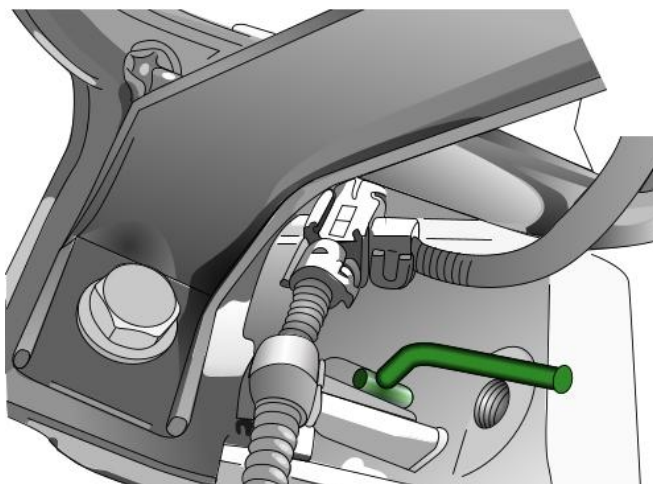


Exhaust:



Lock the flywheel

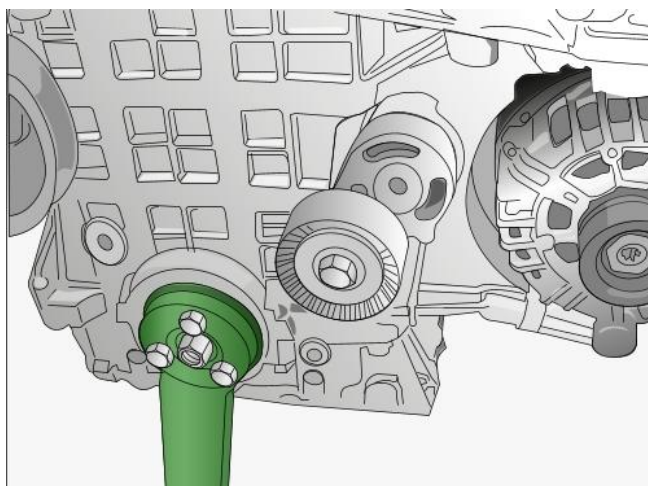
Use the special tool (11 5 120)

Repair Manuals

Fit the special tool: (11 9 280)

Remove the flywheel locking tool (11 5 120)

Loosen the centre bolt of the crankshaft gearwheel/vibration damper (90°)



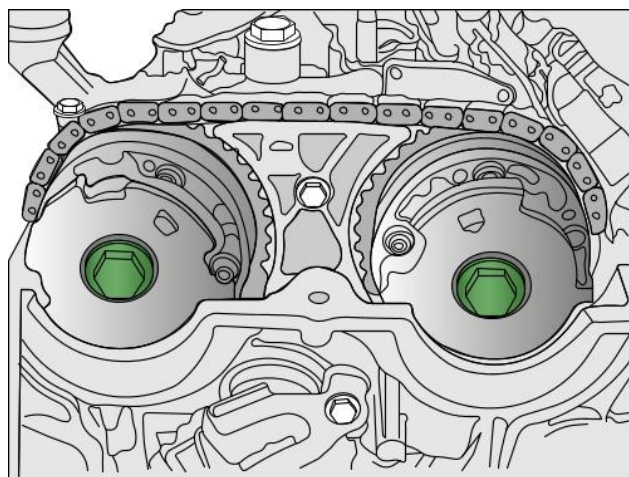
Remove the special tool (11 9 280)

Turn the crankshaft to TDC for cylinder 1

Fit the flywheel locking tool (11 5 120)

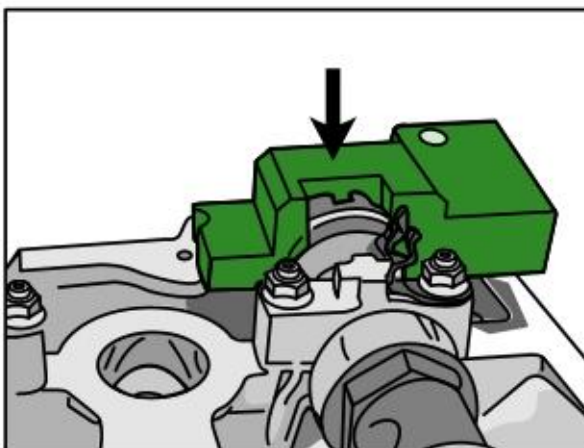
Check that the camshafts are locked

Loosen the camshaft gearwheel bolts

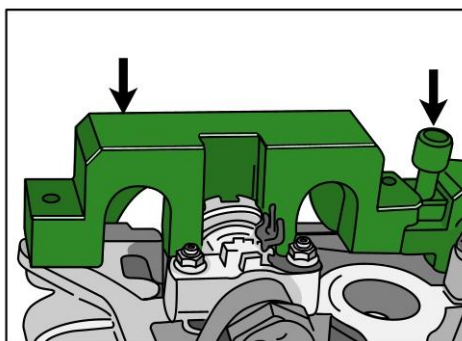


Use the special tools to lock the camshafts (11 7 252)

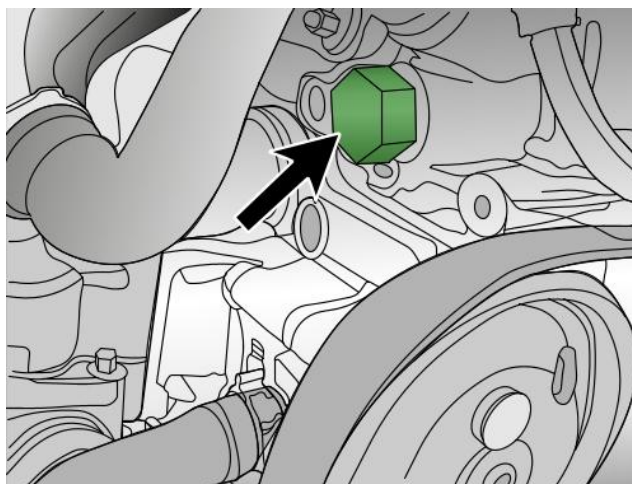
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(11 7 251, 11 7 253)



Remove the chain tensioner piston



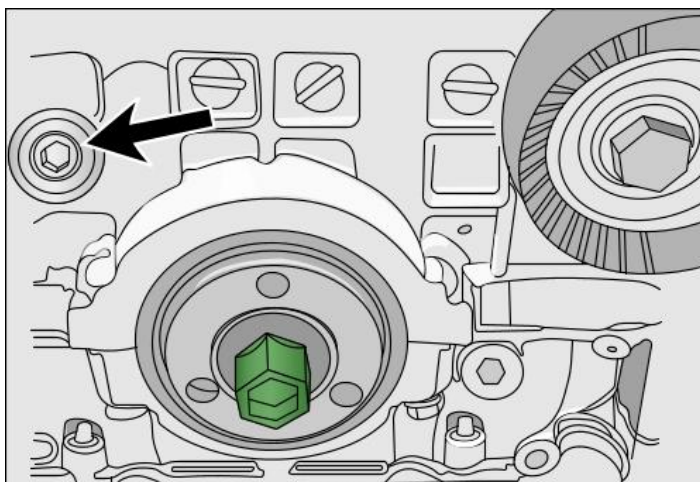
Remove the exhaust camshaft timing adjuster

Remove the inlet camshaft timing adjuster

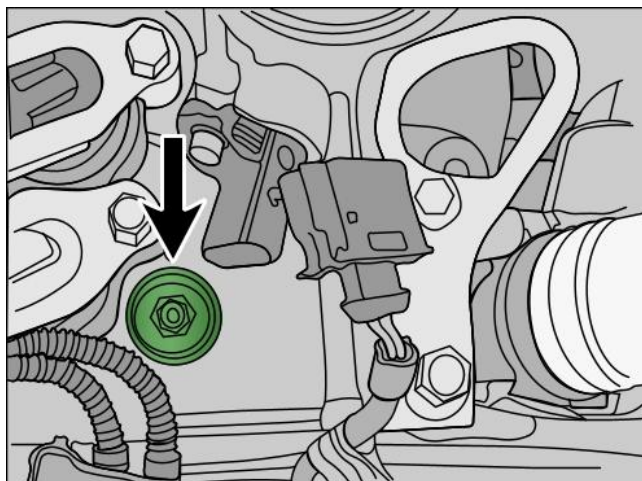
Remove the crankshaft gearwheel centre bolt

Remove the crankshaft hub

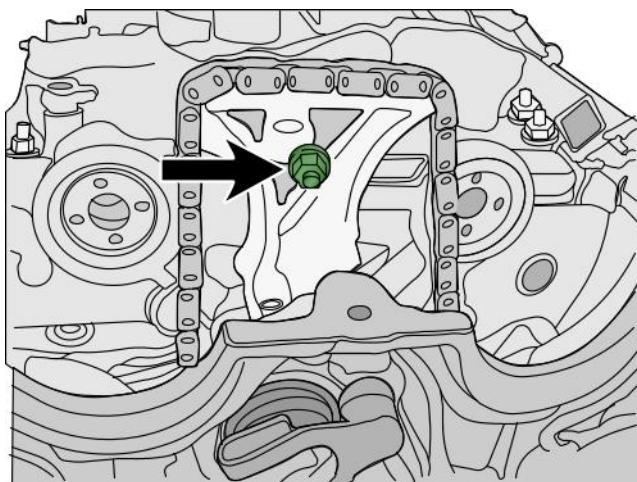
Remove the access plug

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Remove the locking pin
Remove the access plug



Remove the timing chain guide bolts from the cylinder head
Remove the chain guide



Remove the timing chain and one of the gearwheels together

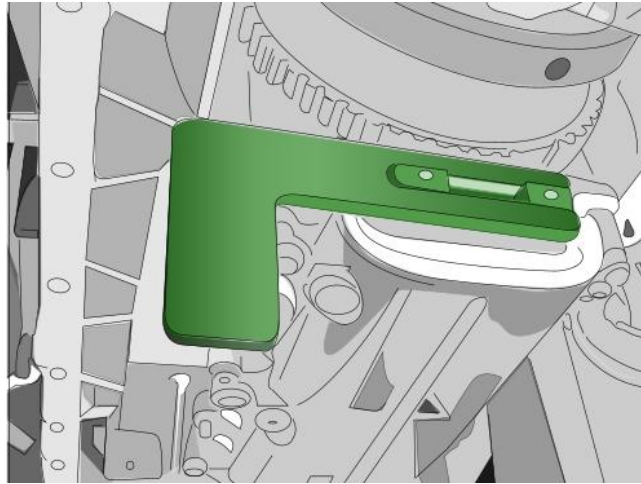
Installation

Fit the timing chain
Fit the locking pin

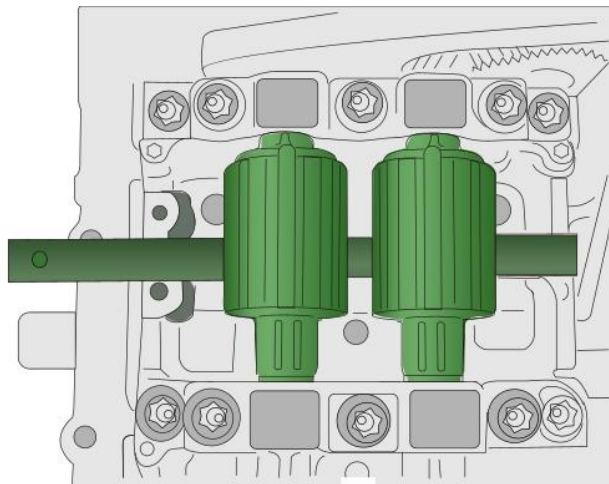
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Refit the access plug (45 Nm)
Refit the crankshaft hub
Check the condition and renew if necessary
Fit the centre bolt of the crankshaft gearwheel/vibration damper
Renew the bolt
Do not tighten!
Align the balance shafts
Lock both balance shafts
Use the special tool

Older type: (11 9 330)

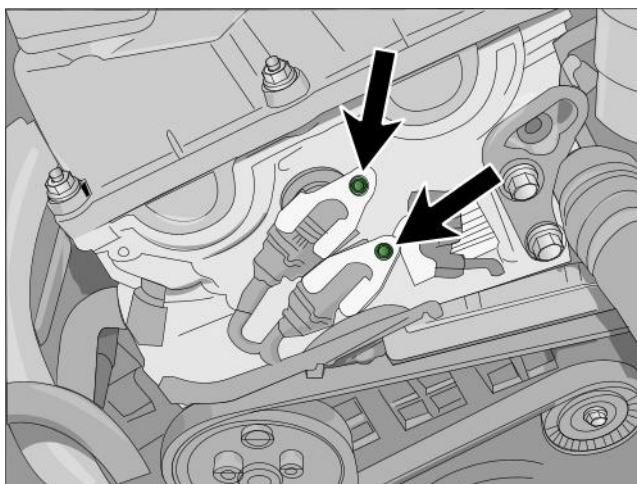


Newer type: (11 5 140)

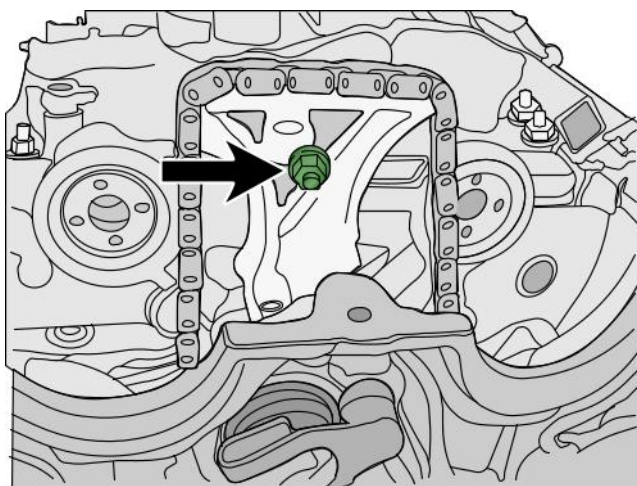


Tighten the crankshaft gearwheel bolt (60 Nm)
Reconnect the connectors to the solenoid valve

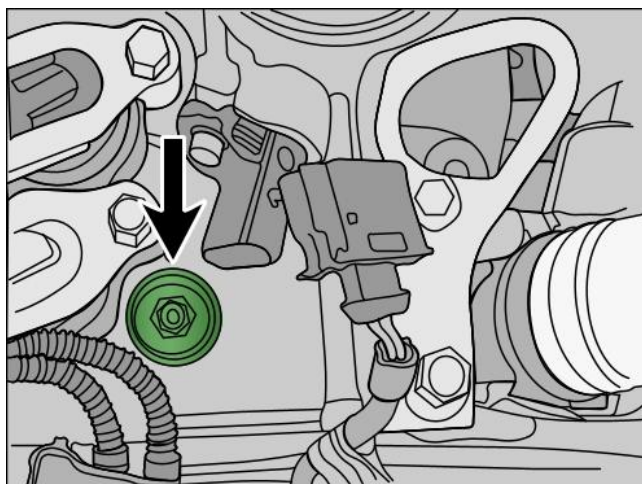
Repair Manuals



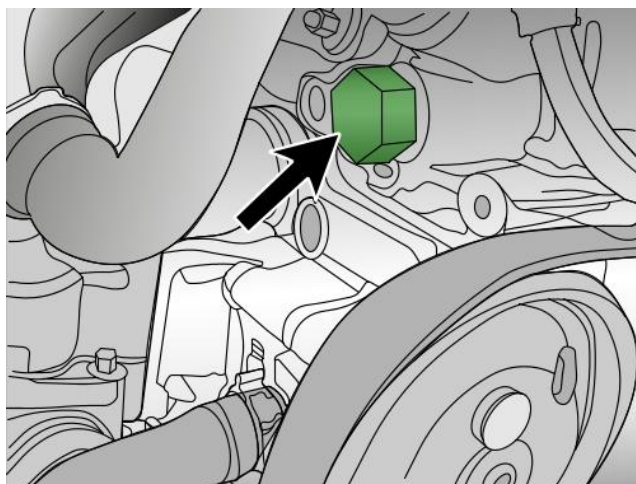
Refit the chain guide



Refit the access plug (25 Nm)



Refit the chain tensioner piston

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Refit the inlet camshaft timing adjuster

Fit the timing chain onto the inlet camshaft gearwheel

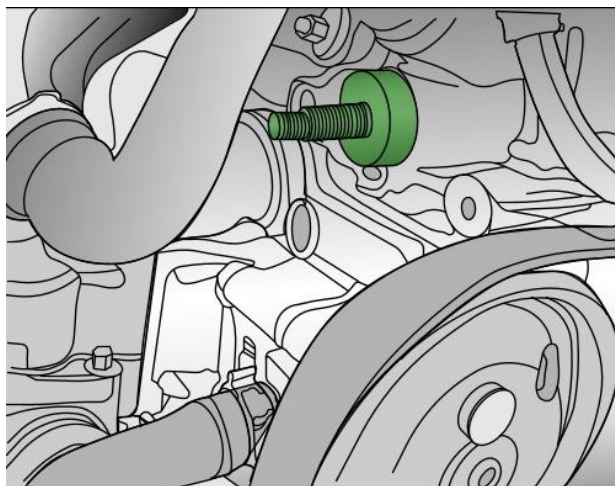
Renew the bolt and tighten

Refit the exhaust camshaft adjuster

Fit the timing chain onto the exhaust camshaft gearwheel

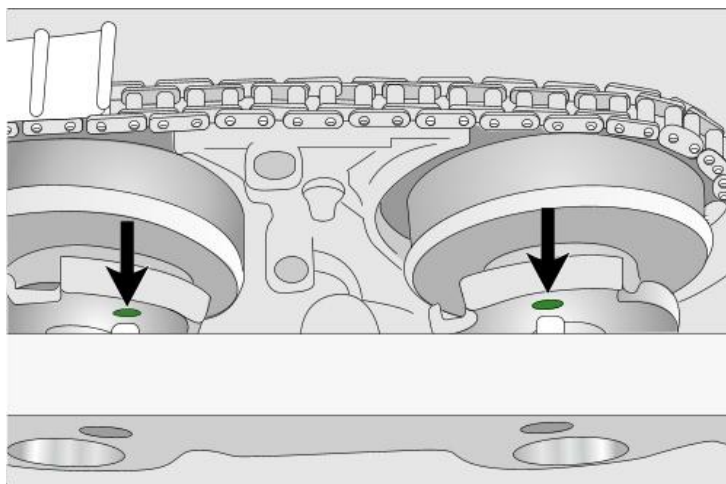
Renew the bolt and tighten

Fit the adjustment tool (11 9 340)



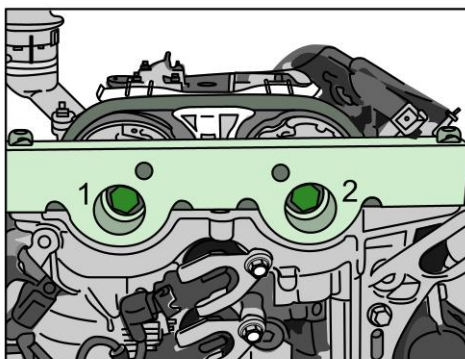
Align the camshaft gearwheel marks as shown

Use the special tool (11 7 260)



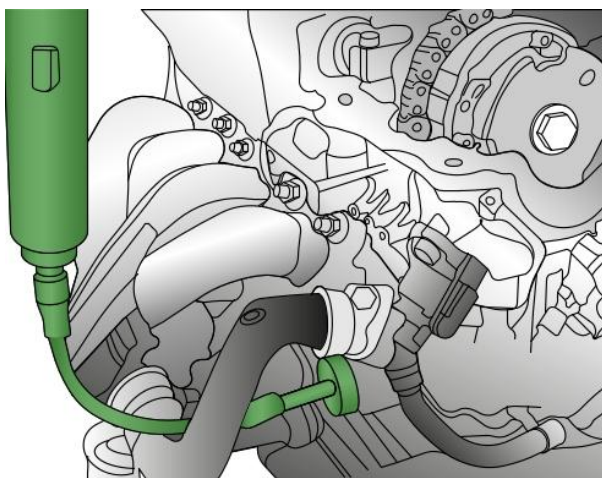
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Loosen the camshaft gearwheel bolts

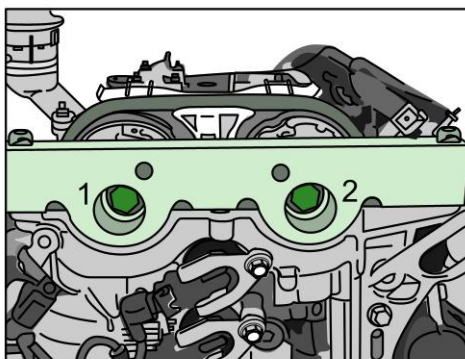


Pre-tension the timing chain

Use the special tool (0.6 Nm) (00 9 250)

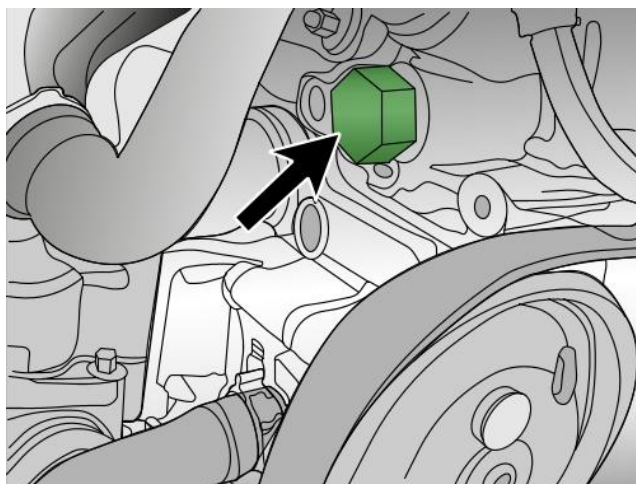


Tighten the camshaft gearwheel bolts



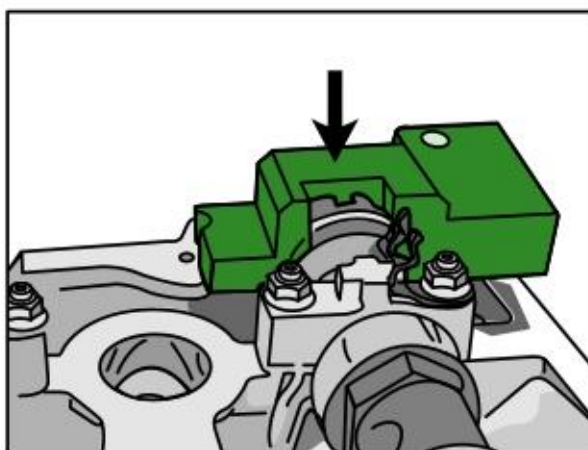
Remove the special tools (11 7 260, 11 9 340)

Refit the chain tensioner piston

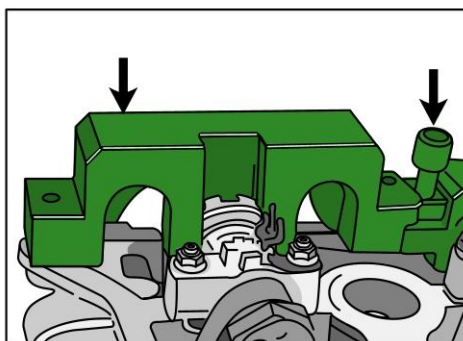
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Check the timing

The timing is set when the special tools fit correctly (11 7 252)



(11 7 251, 11 7 253)



Refit the cylinder head cover

Refit the spark plugs

Refit the ignition coils

Renew the crankshaft front radial seal

Remove the flywheel locking tool (11 5 120)

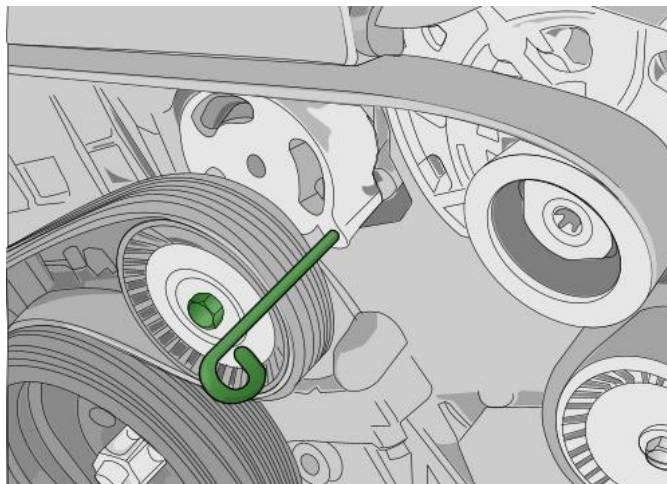
Tighten the crankshaft gearwheel bolt (11 9 280)

Repair Manuals

Fit the vibration damper (34 Nm)

Refit the ancillary drive belt

Remove the tensioner locking tool (11 3 340)



Refit the air filter housing

Refit the mass airflow meter

Reconnect the battery earth cable

Torque settings

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt

Stage 1 50 (Nm)

Stage 2 150 (°)

Crankshaft pulley

Crankshaft pulley (Outer bolts) 34 (Nm)

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 90 (°)

Stage 3 90 (°)

Spark plugs

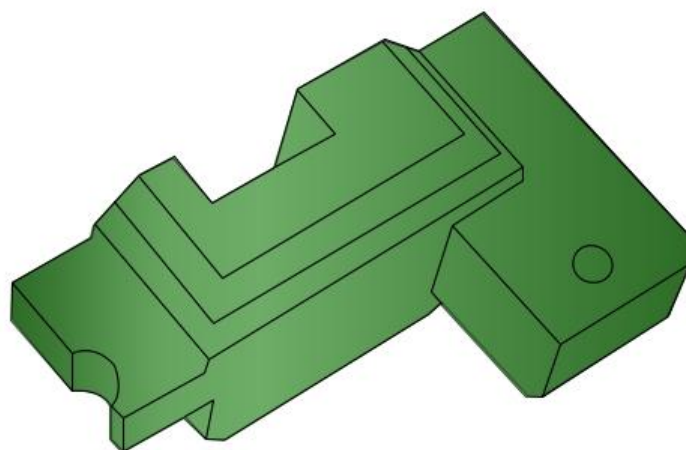
Spark plugs M12: 23 ± 3 (Nm)

Spark plugs M14: 30 ± 3 (Nm)

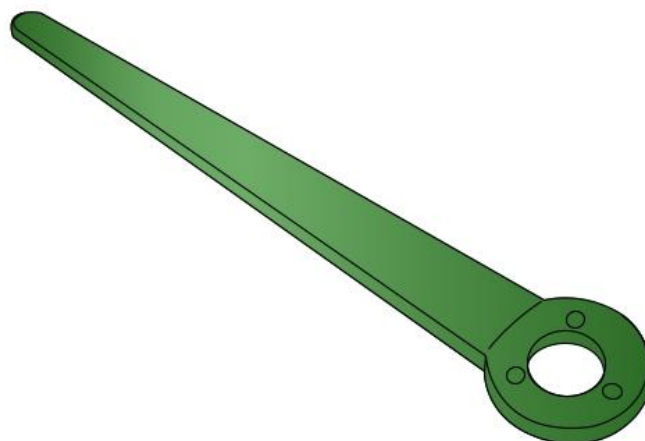
Special tools

Camshaft locking tool: (11 7 252)

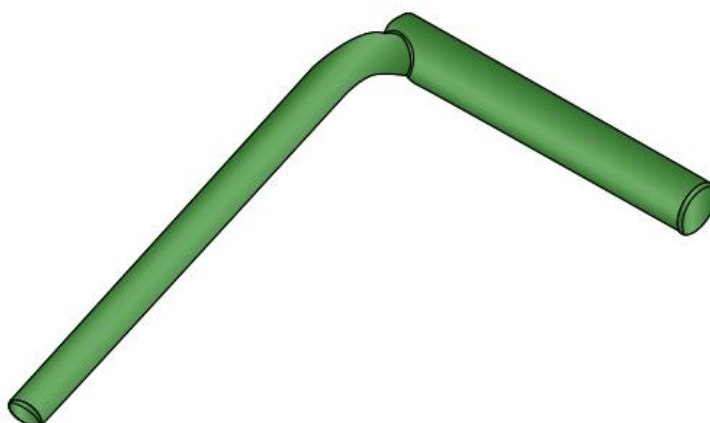
Repair Manuals



Crankshaft centering tool: (11 9 280)

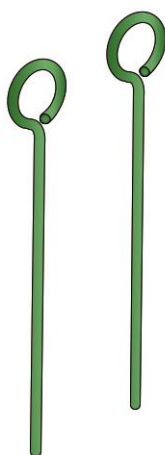


Flywheel locking tool: (11 5 120)

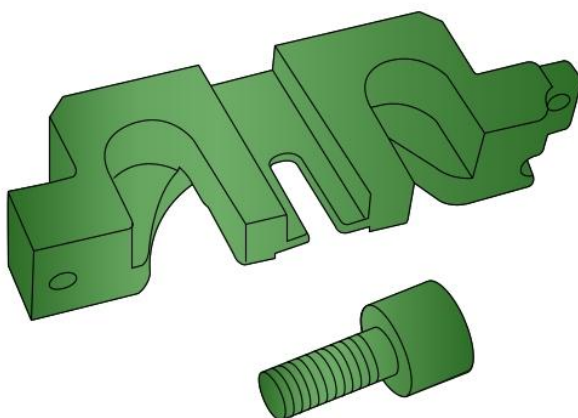


Chain tensioner locking pin: (11 3 340)

Repair Manuals



Camshaft locking tool: (11 7 251, 11 7 253)

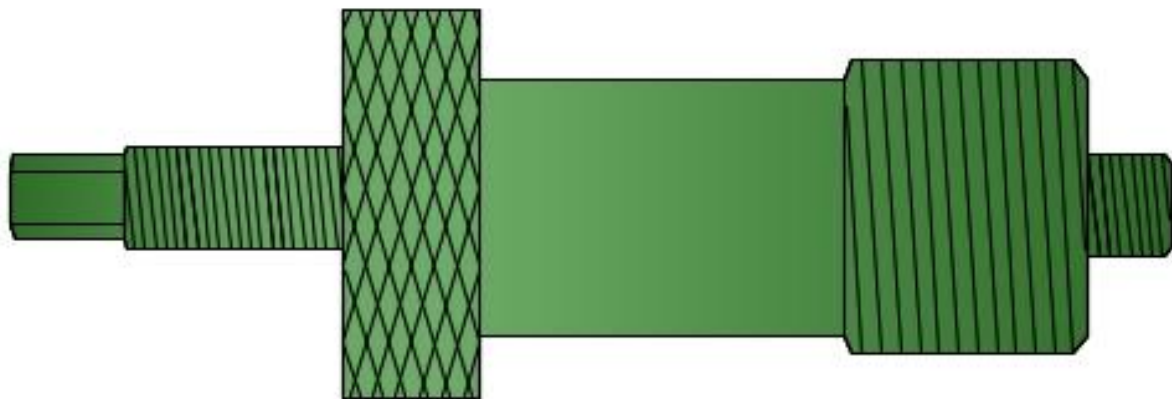


Camshaft alignment tool: (11 7 260)

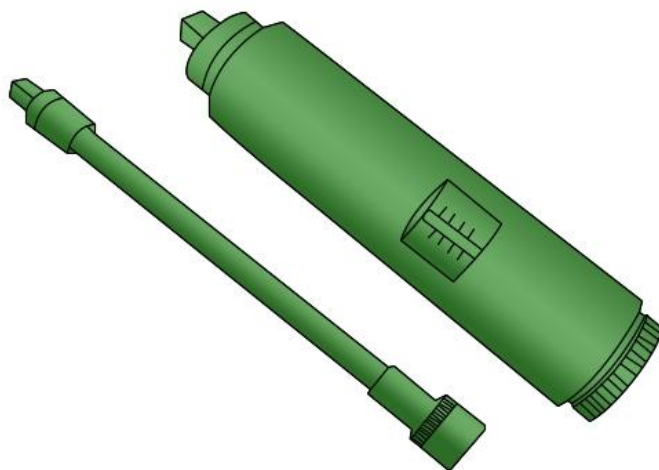


Adaptor: (11 9 340)

Repair Manuals



Adjustment tool: (00 9 250)



Balance shaft tensioning tool: (11 5 140)



Balance shaft tensioning tool: (11 9 330)

Repair Manuals

